

ABSTRACT

This study addresses the dual challenges of operational delays in local public governance and the untracked economic potential of community environmental initiatives. Proponents designed and evaluated Project BIG-ES, a centralized web-based platform tailored for Barangay Bagbag, Quezon City, that unifies public e-services with Material Recovery Facility (MRF) and urban farming monitoring modules. Developed using the Laravel framework, MySQL, and Python, the system replaces manual record-keeping with a secure, semi-automated verification-first workflow that validates resident registrations using document uploads before granting access to document requests. Financial and material logs are consolidated into a centralized ledger to serve as inputs for the integrated Prophet forecasting model, which projects multi-month revenue patterns for administrative budget planning. To bridge user accessibility gaps, the platform features a unique Smart Report function that automatically translates complex time-series charts into plain, text-based narrative summaries. An empirical evaluation with 83 respondents using the ISO/IEC 25010 software quality standard yielded a Grand Mean score of 3.29, indicating an interpretation of "Strongly Agree" and confirming that the platform successfully passed the validation threshold. Barangay admins strongly praised the system's flexibility (3.57), residents highly valued its reliability (3.36), and external IT experts technically validated its functional suitability (3.66) and security (3.60). The results prove that Project BIG-ES effectively modernizes frontline governance workflows while demonstrating the fiscal viability of local sustainability programs.

Keywords: Analytics, Waste Management, Urban Farming, E-services, Smart Report